Fixation probabilities as absorption limits of matrix powers, kernel-checked: neutral Wright–Fisher, the Moran model with selection, the Price equation, and a fail-closed CVODE certificate

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AI-assistance disclosure. This preprint, its Lean code, its Rust and Python code and its claim ledger were produced by AI agents (Claude, Anthropic) under the author's direction. The author is responsible for the content. No human has audited the Lean statements for adequacy, and no human referee has checked the hand proofs. The Lean files were compiled by their producer agents and re-compiled at the paper stage by the same AI pipeline. A separate AI agent then audited the preprint: it recompiled every Lean module from source in its own build directory (not the producer's `.olean` files), ran its own axiom audit and its own planted controls, reproduced the CVODE results byte for byte, re-derived the numbers, hand proofs and citations, and gave the verdict ACCEPT_WITH_CORRECTIONS with no blocking issue (`AUDIT.md`; this version applies its corrections, see `REVISION.md`). That auditor is an AI of the same model family as the producer, so this is an independent recompile and not a human audit. Three things changed after the audit and were re-checked only by the revising agent: a doc comment in `FixationDataCert.lean` (recompiled, rc 0, same axiom lines), the Rust crate (a componentwise non-negativity check, commit `cda9a90`), and, as a consequence, `results.jsonl` (regenerated: one added field and one `null`; the auditor's byte-for-byte reproduction refers to the earlier file, and every other field is unchanged). Treat every Tier A statement below as "kernel-checked, statement adequacy not audited by a human".

Abstract

For a stochastic kernel P on $\{0, \dots, N\}$ with absorbing ends we prove in Lean 4 that the n-step probability of being at N converges to $h(i)$, and that of being at 0 to $1 - h(i)$, whenever h is harmonic at the interior states with boundary values 0 and 1 and the one-step variance of h is positive at every interior state. The proof is a second-moment argument: the interior mass has bounded partial sums, hence tends to 0, and a squeeze finishes. One theorem covers two models. For neutral Wright–Fisher, $P^n(i, N) \to i/N$ (this closes a gap recorded in an earlier module, which proved only uniqueness of the harmonic function). For the Moran process with

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fitness $r > 0$, $P^n(i, N) \to S_i/S_N$ with $S_n = \sum_{k<n} r^{-k}$, that is $(1 - r^{-i})/(1 - r^{-N})$ for $r \neq 1$ and i/N for $r = 1$; the limit $r \to 1$ of the closed form is proved too. We also prove the Price equation for a finite population as an exact identity over any field, and its haploid corollary $\Delta\bar{w} = \mathrm{Var}_n(w)/\bar{w}$ for constant type fitnesses and no transmission bias.

Tier A (Lean 4.34.1, Mathlib only; axioms within `propext`, `Classical.choice`, `Quot.sound`; planted `sorry` and planted-axiom controls caught; kernel-checked positive and negative controls). **What is not proved:** the identification of the matrix power $P^n(i, N)$ with the probability $\mathbb{P}_i(T_N \leq n)$ of a path measure is the standard definition of the n-step law and is not formalised (Tier C); and the preregistered "Fisher's theorem in the partial-change reading" was *not achieved*: only the haploid constant-fitness special case is proved. **Tier X:** a new rusty-SUNDIALS crate, `qf-popgen`, integrates the uniformised Kolmogorov equation with CVODE (BDF, exact Jacobian) behind a *fail-closed* absorption certificate (kernel validity, mass conservation, componentwise non-negativity, decay of the interior mass, agreement with the target): 106 accepted runs agree with the closed forms to $\leq 3.1e-10$ (and with `scipy.linalg.expm` to $\leq 1.0e-13$), and all 9 negative controls are refused. All results are classical; no novelty is claimed, and nothing here depends on openai/math results.

Tiers. [A]: kernel-checked Lean, axioms within the allowlist, planted controls caught. [L]: a cited source, quoted in the ledger (prose as extracted, mathematics transcribed linearly, elisions marked), pinned to a page. [C]: an argument or human proof that is not machine-checked. [X]: floating-point exploration, which never supports a claim. Identifiers such as `PG-A-0001` refer to `ledger/ledger.json` (Section 7). A Tier A tag attaches to the Lean statement, never to its biological gloss. Status words: every module was compiled by its producer, re-compiled by the same AI pipeline, and then *recompiled from source by a separate AI auditor* (own build directory, own axiom audit, own planted controls; verdict ACCEPT_WITH_CORRECTIONS); the one module imported from earlier work (`PopGenWrightFisher.lean`, "H11") was likewise recompiled by a separate AI verifier. No module has had a human audit, and the auditor shares the producer's model family.

1 Setting

A population of N individuals contains i mutants. A *kernel* P is an $(N + 1) \times (N + 1)$ matrix with non-negative entries and unit row sums; 0 *and* N *are absorbing* if rows 0 and N are the point masses δ_0, δ_N. The n-step kernel is defined by the Chapman–Kolmogorov recursion $P^0 = I$, $P^{n+1}(i, j) = \sum_k P^n(i, k)P(k, j)$ (`kPow` in Lean), the interior mass is $I_n(i) = \sum_{0<k<N} P^n(i, k)$, and h is *interior-harmonic* if $\sum_j P(i, j)h(j) = h(i)$ for $0 < i < N$. Listing 1 gives these definitions verbatim up to glyph rendering (the listings replace a few Unicode symbols by LaTeX glyphs, the symbols for the reals, the neighbourhood filter and negation appear as $\mathbb{R}$, $\mathcal{N}$, $\neg$; the Lean files in the repository are the reference).

Listing 1: Chains on `Fin (N+1)`: definitions (`Absorption.lean`, verbatim up to glyph rendering).

```
noncomputable def kPow (P : Fin (N + 1) → Fin (N + 1) → ℝ) : ℕ → Fin (N + 1) → Fin (N + 1) → ℝ
  | 0 => fun i j => if i = j then 1 else 0
  | n + 1 => fun i j => ∑ k, kPow P n i k * P k j

structure AbsorbingChain (P : Fin (N + 1) → Fin (N + 1) → ℝ) : Prop where
  nonneg : ∀ i j, 0 ≤ P i j
  rowSum : ∀ i, ∑ j, P i j = 1
  absorb0 : ∀ j, P 0 j = if j = 0 then 1 else 0
  absorbN : ∀ j, P (Fin.last N) j = if j = Fin.last N then 1 else 0

def InteriorHarmonic (P : Fin (N + 1) → Fin (N + 1) → ℝ) (g : Fin (N + 1) → ℝ) : Prop :=
  ∀ i : Fin (N + 1), 0 < (i : ℕ) → (i : ℕ) < N → ∑ j, P i j * g j = g i
```

The two models are the neutral Wright–Fisher kernel $\mathrm{wfP}_N(i,j) = \binom{N}{j}(i/N)^j(1-i/N)^{N-j}$ (imported unchanged from the earlier module H11) and the Moran process with fitness $r > 0$: with i mutants, one individual is chosen to reproduce with probability proportional to fitness and one uniformly at random dies, so the mutant count moves up with probability p_i, down with probability q_i and stays otherwise,

$$p_i = \frac{ri}{ri + N - i} \cdot \frac{N-i}{N}, \qquad q_i = \frac{N-i}{ri + N - i} \cdot \frac{i}{N}.$$

Keller and Uğurlu write the same rates as $p^+(j) = \frac{r}{1+(r-1)j/n}\frac{j}{n}\frac{n-j}{n}$ and $p^-(j) = \frac{1}{1+(r-1)j/n}\frac{n-j}{n}\frac{j}{n}$, "the individual to be replaced [being] chosen uniformly at random" [L, PG-L-0001]; the equality is a one-line computation, $\frac{rj/n}{1+(r-1)j/n} = \frac{rj}{rj+n-j}$ [C, PG-C-0004]. Their fixation probabilities are $\varrho_i = i/n$ ($r = 1$) and $(1-1/r^i)/(1-1/r^n)$ ($r \neq 1$), derived by optional stopping on the martingale r^{-M_k} [L, PG-L-0002]. Our route avoids a path measure and works on matrix powers (Listing 2).

Listing 2: The Moran model and its fixation function (`MoranModel.lean`, verbatim up to glyph rendering).

```
noncomputable def bdP (N : ℕ) (p q : ℕ → ℝ) (i j : Fin (N + 1)) : ℝ :=
  (if (j : ℕ) = (i : ℕ) + 1 then p i else 0) + (if (j : ℕ) + 1 = (i : ℕ) then q i else 0) +
    (if (j : ℕ) = (i : ℕ) then 1 - p i - q i else 0)

noncomputable def moranUp (N : ℕ) (r : ℝ) (i : ℕ) : ℝ :=
  r * i / (r * i + ((N : ℝ) - i)) * (((N : ℝ) - i) / N)

noncomputable def moranDown (N : ℕ) (r : ℝ) (i : ℕ) : ℝ :=
  ((N : ℝ) - i) / (r * i + ((N : ℝ) - i)) * ((i : ℝ) / N)

noncomputable def moranP (N : ℕ) (r : ℝ) : Fin (N + 1) → Fin (N + 1) → ℝ :=
  bdP N (moranUp N r) (moranDown N r)

noncomputable def geomS (r : ℝ) (n : ℕ) : ℝ := ∑ k ∈ range n, r⁻¹ ^ k

noncomputable def moranFix (N : ℕ) (r : ℝ) (i : ℕ) : ℝ := geomS r i / geomS r N
```

Bridge to probability (Tier C, PG-C-0001). $P^n(i,N)$ is by definition the n-step transition probability $\mathbb{P}_i(X_n = N)$ of the chain with kernel P; since N is absorbing, $\{X_n = N\} = \{T_N \le n\}$, so $P^n(i,N) = \mathbb{P}_i(T_N \le n)$, which is consistent with the kernel-checked monotonicity of $P^n(i,N)$ in n. The limit is therefore the fixation probability $\mathbb{P}_i(T_N < \infty)$. This identification is standard and is *not* a Lean statement here: no path space is built. Mathlib has an Ionescu–Tulcea construction, `ProbabilityTheory.Kernel.traj`, which Marion describes as integrated into Mathlib [L, PG-L-0003]; that it is present in the pinned build is a repository fact (declared in `Mathlib/Probability/Kernel/IonescuTulcea/Traj.lean`, line 518, found by grep in the audit), not a citation. It could carry the bridge; we do not use it. Wording used throughout: "the n-step absorption probability at N converges to…; kernel-checked".

2 Step-by-step human demonstration

This section is for a strong undergraduate. Each step states a fact, proves it by hand [C, PG-C-0003], and names the Lean declaration that checks the matching statement [A]. Let P be a kernel on $\{0,\dots,N\}$, $N > 0$, with $0, N$ absorbing, and let h be interior-harmonic with $h(0) = 0$, $h(N) = 1$ and

$$v(k) = \sum_j P(k,j)\big(h(j) - h(k)\big)^2 > 0 \quad (0 < k < N).$$

Step 1: martingale identity [A, PG-A-0001]. *Claim:* $P^n h = h$ for all n, i.e. $\sum_j P^n(i,j)h(j) = h(i)$. *Proof.* At an absorbing end the row is a point mass, so $(Ph)(0) = h(0)$ and $(Ph)(N) = h(N)$; with interior harmonicity, $Ph = h$ everywhere. Then $P^{n+1}h = P^n(Ph) = P^n h$, and induction on n. The same argument with $h \equiv 1$ shows that rows of P^n sum to 1; non-negativity of P^n is induction on the recursion. Lean: `kPow_harmonic`, `kPow_rowSum`, `kPow_nonneg`.

Step 2: second-moment identity [A, PG-A-0001]. *Claim:* $\sum_j P(k,j)h(j)^2 = h(k)^2 + v(k)$. *Proof.* Expand $(h(j) - h(k))^2 = h(j)^2 - 2h(k)h(j) + h(k)^2$ and use $\sum_j P(k,j) = 1$ and $\sum_j P(k,j)h(j) = h(k)$. Lean: `second_moment`.

Step 3: the interior mass is summable [A, PG-A-0001]. *Claim:* $I_n(i) \to 0$. *Proof.* Let $\varepsilon = \min_{0<k<N} v(k) > 0$ (a minimum over finitely many positive numbers). By Step 2, pointwise $P(h^2) \geq h^2 + \varepsilon\, \mathbf{1}_{\text{int}}$: at interior k this is $v(k) \geq \varepsilon$, and at an end state both sides equal $h(k)^2$. Apply P^m, whose entries are ≥ 0: $P^{m+1}(h^2) \geq P^m(h^2) + \varepsilon I_m$. Summing over $m < n$,

$$\varepsilon \sum_{m<n} I_m(i) \leq P^n(h^2)(i) - h(i)^2 \leq \sum_j h(j)^2 =: G,$$

since the rows of P^n sum to 1. The partial sums of the non-negative series $\sum_m I_m(i)$ are bounded by G/ε, so it converges and its terms tend to 0. Lean: `interiorMass_tendsto_zero`.

Step 4: the squeeze [A, PG-A-0001]. *Claim:* $P^n(i,N) \to h(i)$ and $P^n(i,0) \to 1 - h(i)$. *Proof.* Put $H = \sum_j |h(j)|$. For every state j, $|h(j) - \mathbf{1}_{j=N}| \leq H\mathbf{1}_{\text{int}}(j)$: at 0 and N the left side is 0; at an interior state it is at most $|h(j)| \leq H$. Applying $P^n(i,\cdot)$ (non-negative weights) and Step 1, $|h(i) - P^n(i,N)| \leq H\, I_n(i) \to 0$. Finally $P^n(i,0) + P^n(i,N) + I_n(i) = 1$ (Lean: `mass_split`), so $P^n(i,0) = 1 - P^n(i,N) - I_n(i) \to 1 - h(i)$. Lean: `absorption_tendsto`, `fixation_limit`.

Step 5: uniqueness, and monotonicity [A, PG-A-0001]. If g is another interior-harmonic function with $g(0) = 0$, $g(N) = 1$, then Steps 1 and 4 apply to g with the *same* $I_n \to 0$ (it depends on h only through $v > 0$), so $P^n(i,N) \to g(i)$ as well; limits are unique, hence $g = h$. This replaces the maximum principle used for Wright–Fisher in H11. And $P^{n+1}(i,N) = \sum_k P^n(i,k)P(k,N) \geq P^n(i,N)P(N,N) = P^n(i,N)$. Lean: `harmonic_unique_of_var`, `kPow_last_mono`.

Listing 3: The general theorem and its main lemmas (`Absorption.lean`, statements verbatim up to glyph rendering, proofs elided).

```
theorem kPow_harmonic (P : Fin (N + 1) → Fin (N + 1) → ℝ) (g : Fin (N + 1) → ℝ)
    (hg : ∀ k, ∑ j, P k j * g j = g k) : ∀ n i, ∑ j, kPow P n i j * g j = g i := -- proof elided

theorem second_moment {P : Fin (N + 1) → Fin (N + 1) → ℝ} (hP : AbsorbingChain P)
    {h : Fin (N + 1) → ℝ} (hall : ∀ k, ∑ j, P k j * h j = h k) (k : Fin (N + 1)) :
    ∑ j, P k j * h j ^ 2 = h k ^ 2 + ∑ j, P k j * (h j - h k) ^ 2 := -- proof elided

theorem interiorMass_tendsto_zero {P : Fin (N + 1) → Fin (N + 1) → ℝ} (hP : AbsorbingChain P)
    {h : Fin (N + 1) → ℝ} (hh : InteriorHarmonic P h)
    (hvar : ∀ k : Fin (N + 1), 0 < (k : ℕ) → (k : ℕ) < N → 0 < ∑ j, P k j * (h j - h k) ^ 2)
    (i : Fin (N + 1)) : Tendsto (fun n => interiorMass P n i) atTop (𝓝 0) := -- proof elided

theorem absorption_tendsto {P : Fin (N + 1) → Fin (N + 1) → ℝ} (hP : AbsorbingChain P)
    (hN : 0 < N) {g : Fin (N + 1) → ℝ} (hg : InteriorHarmonic P g) (g0 : g 0 = 0)
    (g1 : g (Fin.last N) = 1)
    (hI : ∀ i, Tendsto (fun n => interiorMass P n i) atTop (𝓝 0)) (i : Fin (N + 1)) :
    Tendsto (fun n => kPow P n i (Fin.last N)) atTop (𝓝 (g i)) := -- proof elided
```

```
theorem fixation_limit {P : Fin (N + 1) → Fin (N + 1) → ℝ} (hP : AbsorbingChain P) (hN : 0 < N)
    {h : Fin (N + 1) → ℝ} (hh : InteriorHarmonic P h) (h0 : h 0 = 0) (h1 : h (Fin.last N) = 1)
    (hvar : ∀ k : Fin (N + 1), 0 < (k : ℕ) → (k : ℕ) < N → 0 < ∑ j, P k j * (h j - h k) ^ 2)
    (i : Fin (N + 1)) :
    Tendsto (fun n => kPow P n i (Fin.last N)) atTop (𝓝 (h i)) ∧
    Tendsto (fun n => kPow P n i 0) atTop (𝓝 (1 - h i)) := -- proof elided

theorem harmonic_unique_of_var {P : Fin (N + 1) → Fin (N + 1) → ℝ} (hP : AbsorbingChain P)
    (hN : 0 < N) {h : Fin (N + 1) → ℝ} (hh : InteriorHarmonic P h) (h0 : h 0 = 0)
    (h1 : h (Fin.last N) = 1)
    (hvar : ∀ k : Fin (N + 1), 0 < (k : ℕ) → (k : ℕ) < N → 0 < ∑ j, P k j * (h j - h k) ^ 2)
    {g : Fin (N + 1) → ℝ} (hg : InteriorHarmonic P g) (g0 : g 0 = 0) (g1 : g (Fin.last N) = 1) :
    ∀ i, g i = h i := -- proof elided
```

Step 6: neutral Wright–Fisher [A, PG-A-0002]. Take $h(j) = j/N$. It is harmonic because the binomial mean is $\sum_j \binom{N}{j} x^j (1-x)^{N-j}\, j/N = x$ at $x = k/N$ (H11's `frac_harmonic`). At interior k, the jump to 0 alone contributes $P(k,0)(k/N)^2 = (1-k/N)^N (k/N)^2 > 0$ to $v(k)$, so $v(k) > 0$. Steps 1–5 give $P^n(i,N) \to i/N$ and $P^n(i,0) \to 1 - i/N$.

Listing 4: Wright–Fisher: variance and the limit (`WrightFisherFixation.lean`, statements verbatim up to glyph rendering).

```
theorem wf_var_pos {N : ℕ} (hN : 0 < N) (k : Fin (N + 1)) (hk1 : 0 < (k : ℕ)) (hk2 : (k : ℕ) < N) :
    0 < ∑ j, wfP N k j * (((j : ℕ) : ℝ) / N - ((k : ℕ) : ℝ) / N) ^ 2 := -- proof elided

theorem wright_fisher_fixation_limit {N : ℕ} (hN : 0 < N) (i : Fin (N + 1)) :
    Tendsto (fun n => kPow (wfP N) n i (Fin.last N)) atTop (𝓝 (((i : ℕ) : ℝ) / N)) ∧
    Tendsto (fun n => kPow (wfP N) n i 0) atTop (𝓝 (1 - ((i : ℕ) : ℝ) / N)) := -- proof elided
```

Step 7: the Moran kernel is an absorbing chain [A, PG-A-0003, PG-A-0004]. The ratio of the rates is $p_i/q_i = r$ (`moranUp_eq`). The stay probability is $1 - p_i - q_i = \frac{ri^2 + (N-i)^2}{(ri+N-i)N} \geq 0$ (`moran_stay_nonneg`); so P is stochastic, and $p_0 = q_0 = p_N = q_N = 0$ makes $0, N$ absorbing.

Step 8: the first-step equation [A, PG-A-0003]. Let $S_n = \sum_{k<n} r^{-k}$ and $h(i) = S_i/S_N$. Then $h(i+1) - h(i) = r^{-i}/S_N$. The interior harmonic equation $p_i h(i+1) + q_i h(i-1) + (1-p_i-q_i)h(i) = h(i)$ is equivalent to $p_i\,(h(i+1) - h(i)) = q_i\,(h(i) - h(i-1))$, i.e. $p_i\, r^{-i} = q_i\, r^{-(i-1)}$, which is $p_i = r q_i$. Lean: `moranFix_harmonic_nat`. The one-step variance at interior i is at least $p_i (r^{-i}/S_N)^2 > 0$ because $p_i > 0$ for $0 < i < N$ (`moran_var_pos_nat`). Steps 1–5 give, for every $r > 0$ and $0 < N$ (`moran_fixation_limit`, `moran_fixation_unique`),

$$P^n(i,N) \longrightarrow \frac{S_i}{S_N}, \qquad P^n(i,0) \longrightarrow 1 - \frac{S_i}{S_N}.$$

Step 9: closed forms and the neutral limit [A, PG-A-0003, PG-A-0004]. For $r \neq 1$, with $\rho = 1/r \neq 1$, the geometric sum gives $S_n = (1-\rho^n)/(1-\rho)$ and so $S_i/S_N = (1-r^{-i})/(1-r^{-N})$ (`moranFix_closed`, `moran_fixation_limit_closed`). For $r = 1$, $S_n = n$ and the value is i/N (`moranFix_neutral`, `moran_fixation_limit_neutral`); this also shows that neutral Moran and neutral Wright–Fisher have the same limit although they are different chains. The closed form is continuous at $r = 1$: each $S_n(r)$ is a finite sum of continuous functions with $S_N(1) = N > 0$, so $(1-r^{-i})/(1-r^{-N}) \to i/N$ as $r \to 1$, $r \neq 1$ (`moran_closed_tendsto_neutral`). That settles the "neutral limit" as a proof rather than a spot check.

Listing 5: Moran lemmas (`MoranModel.lean`, statements verbatim up to glyph rendering).

```
theorem moranFix_harmonic_nat {N : ℕ} {r : ℝ} (hr : 0 < r) {i : ℕ} (h1 : 0 < i) :
    moranUp N r i * moranFix N r (i + 1) + moranDown N r i * moranFix N r (i - 1)
      + (1 - moranUp N r i - moranDown N r i) * moranFix N r i = moranFix N r i := -- proof elided

theorem moran_var_pos_nat {N : ℕ} {r : ℝ} (hN : 0 < N) (hr : 0 < r) {i : ℕ} (h1 : 0 < i)
    (h2 : i < N) :
    0 < moranUp N r i * (moranFix N r (i + 1) - moranFix N r i) ^ 2
      + moranDown N r i * (moranFix N r (i - 1) - moranFix N r i) ^ 2
      + (1 - moranUp N r i - moranDown N r i) * (moranFix N r i - moranFix N r i) ^ 2 := -- proof elided

theorem moranFix_closed {N : ℕ} {r : ℝ} (hr1 : r ≠ 1) (i : ℕ) :
    moranFix N r i = (1 - r⁻¹ ^ i) / (1 - r⁻¹ ^ N) := -- proof elided

theorem moranFix_neutral (N i : ℕ) : moranFix N 1 i = (i : ℝ) / N := -- proof elided

theorem moran_closed_tendsto_neutral {N : ℕ} (hN : 0 < N) (i : ℕ) :
    Tendsto (fun r : ℝ => (1 - r⁻¹ ^ i) / (1 - r⁻¹ ^ N)) (𝓝[≠] 1) (𝓝 ((i : ℝ) / N)) := -- proof elided
```

Listing 6: Moran chain theorems (`MoranFixation.lean`, statements verbatim up to glyph rendering).

```
theorem moran_fixation_limit {N : ℕ} {r : ℝ} (hN : 0 < N) (hr : 0 < r) (i : Fin (N + 1)) :
    Tendsto (fun n => kPow (moranP N r) n i (Fin.last N)) atTop (𝓝 (moranFix N r i)) ∧
    Tendsto (fun n => kPow (moranP N r) n i 0) atTop (𝓝 (1 - moranFix N r i)) := -- proof elided

theorem moran_fixation_limit_closed {N : ℕ} {r : ℝ} (hN : 0 < N) (hr : 0 < r) (hr1 : r ≠ 1)
    (i : Fin (N + 1)) :
    Tendsto (fun n => kPow (moranP N r) n i (Fin.last N)) atTop
      (𝓝 ((1 - r⁻¹ ^ (i : ℕ)) / (1 - r⁻¹ ^ N))) := -- proof elided

theorem moran_fixation_limit_neutral {N : ℕ} (hN : 0 < N) (i : Fin (N + 1)) :
    Tendsto (fun n => kPow (moranP N 1) n i (Fin.last N)) atTop (𝓝 (((i : ℕ) : ℝ) / N)) := -- proof elided

theorem moran_fixation_unique {N : ℕ} {r : ℝ} (hN : 0 < N) (hr : 0 < r) {g : Fin (N + 1) → ℝ}
    (hg : InteriorHarmonic (moranP N r) g) (g0 : g 0 = 0) (g1 : g (Fin.last N) = 1) :
    ∀ i, g i = moranFix N r i := -- proof elided
```

Step 10: the Price equation [A, PG-A-0005]. A finite population has types $i \in s$ with abundance n_i, fitness w_i (offspring per individual), parental trait z_i and mean offspring trait z'_i. Write $M = \sum n_i$, $A = \sum n_i w_i$, $B = \sum n_i z_i$, $C = \sum n_i w_i z_i$, $D = \sum n_i w_i z'_i$. Then $\bar{w} = A/M$, $\bar{z} = B/M$ and the offspring-generation mean is $\bar{z}' = D/A$; this definition is the modelling assumption. Moreover $\mathrm{Cov}_n(w, z) = \frac{1}{M}\sum n_i(w_i - \bar{w})(z_i - \bar{z}) = (C - AB/M)/M$ and $\mathbb{E}_n[w\,\Delta z] = (D - C)/M$. Hence, if $M \neq 0$ and $A \neq 0$,

$$\bar{w}\,(\bar{z}' - \bar{z}) = \frac{A}{M}\Big(\frac{D}{A} - \frac{B}{M}\Big) = \frac{D}{M} - \frac{AB}{M^2} = \mathrm{Cov}_n(w, z) + \mathbb{E}_n[w\,\Delta z].$$

Lean: `price_equation` (any field, any `Finset` of types; proof: expand the two numerators into the five raw sums, then `field_simp; ring`), `wcov_eq_moment`, and `price_equation_pos` over $\mathbb{R}$ under $s \neq \emptyset$, $n_i > 0$, $w_i > 0$.

Step 11: the haploid selection increment [A, PG-A-0006]. Set the trait equal to fitness, $z = w$, and assume no transmission bias, $z' = w$ (clonal types, constant fitnesses, no mutation). The transmission term $\mathbb{E}_n[w(w - w)]$ is identically 0, so $\bar{w}(\bar{w}' - \bar{w}) = \mathrm{Var}_n(w)$, i.e. $\Delta\bar{w} = \mathrm{Var}_n(w)/\bar{w} \geq 0$ (`fisher_haploid`, `fisher_haploid_div`, `fisher_haploid_nondecreasing`, `fisher_haploid_pos`). This is LaGatta's eq. (2.27) with its environmental term set to zero [L, PG-L-0004], and the discrete analogue of Baez's remark that $d\bar{f}/dt = \mathrm{Var}(f)$ when the fitness functions are constant [L, PG-L-0005].

Remark 1 (Scope of Step 11, Tier C, PG-C-0005). This is *not* Fisher's fundamental theorem in the Price (1972)/Ewens/Edwards partial-change reading, which concerns the change "due to natural selection" with additive genetic variance, and it is not the diploid multi-allele case. Price's statement that the theorem is about the fraction of the change due to selection is quoted by LaGatta (p. 4); Baez records that the naive reading holds only under very restrictive conditions. The preregistered item `fisher_general` (prior 0.35) was **not achieved**; negative results are reported.

Listing 7: Price: definitions (`Price.lean`, verbatim up to glyph rendering).

```
def wmean (s : Finset ι) (n f : ι → K) : K :=
  (∑ i ∈ s, n i * f i) / ∑ i ∈ s, n i

def wcov (s : Finset ι) (n f g : ι → K) : K :=
  (∑ i ∈ s, n i * (f i - wmean s n f) * (g i - wmean s n g)) / ∑ i ∈ s, n i

def wvar (s : Finset ι) (n f : ι → K) : K :=
  wcov s n f f

def zbarNext (s : Finset ι) (n w z' : ι → K) : K :=
  wmean s (fun i => n i * w i) z'

def wbarNext (s : Finset ι) (n w : ι → K) : K :=
  zbarNext s n w w
```

Listing 8: Price equation and the haploid increment (statements verbatim up to glyph rendering, proofs elided).

```
theorem price_equation (s : Finset ι) (n w z z' : ι → K)
    (hN : ∑ i ∈ s, n i ≠ 0) (hW : ∑ i ∈ s, n i * w i ≠ 0) :
    wmean s n w * (zbarNext s n w z' - wmean s n z)
      = wcov s n w z + wmean s n (fun i => w i * (z' i - z i)) := -- proof elided

theorem wcov_eq_moment (s : Finset ι) (n f g : ι → K) (hN : ∑ i ∈ s, n i ≠ 0) :
    wcov s n f g = wmean s n (fun i => f i * g i) - wmean s n f * wmean s n g := -- proof elided

theorem fisher_haploid (s : Finset ι) (n w : ι → K)
    (hN : ∑ i ∈ s, n i ≠ 0) (hW : ∑ i ∈ s, n i * w i ≠ 0) :
    wmean s n w * (wbarNext s n w - wmean s n w) = wvar s n w := -- proof elided

theorem fisher_haploid_div (s : Finset ι) (n w : ι → K)
    (hN : ∑ i ∈ s, n i ≠ 0) (hW : ∑ i ∈ s, n i * w i ≠ 0) :
    wbarNext s n w - wmean s n w = wvar s n w / wmean s n w := -- proof elided

theorem fisher_haploid_nondecreasing (s : Finset ι) (n w : ι → ℝ)
    (hn : ∀ i ∈ s, 0 ≤ n i) (hN : 0 < ∑ i ∈ s, n i) (hW : 0 < ∑ i ∈ s, n i * w i) :
    wmean s n w ≤ wbarNext s n w := -- proof elided

theorem price_equation_pos (s : Finset ι) (n w z z' : ι → ℝ) (hs : s.Nonempty)
    (hn : ∀ i ∈ s, 0 < n i) (hw : ∀ i ∈ s, 0 < w i) :
    wmean s n w * (zbarNext s n w z' - wmean s n z)
      = wcov s n w z + wmean s n (fun i => w i * (z' i - z i)) := -- proof elided
```

Crosswalk to the Genetic repository [A, PG-A-0007; C, PG-C-0007]. The module `ProgPopGen.Price.Alix` copies `meanFit` and `selectP` verbatim from `SocrateAI-Scientific-Genetic/formal/ANSE/AlixPopulationGenetics.lean` and restates two of its three H12 theorems with identical statement text (Listing 9): the first is *derived from the multi-type* `fisher_haploid_div` at `Fin 2`, the second is the `wvar` definition at `Fin 2`. The match is textual only: no Lean term links the two namespaces, and the module was compiled only under the AutoevolveAI pin, not inside the Genetic repository's own build. That repository contains no Price-equation result (whole-repository search; absence by search).

Listing 9: Crosswalk statements (`Price.lean`, namespace `ProgPopGen.Price.Alix`, verbatim up to glyph rendering).

```
def meanFit (w₁ w₂ p : ℚ) : ℚ := p * w₁ + (1 - p) * w₂

def selectP (w₁ w₂ p : ℚ) : ℚ := p * w₁ / meanFit w₁ w₂ p

theorem aurelia_mean_fitness_increment (w₁ w₂ p : ℚ) (hm : meanFit w₁ w₂ p ≠ 0) :
    meanFit w₁ w₂ (selectP w₁ w₂ p) - meanFit w₁ w₂ p
      = p * (1 - p) * (w₁ - w₂) ^ 2 / meanFit w₁ w₂ p := -- proof elided

theorem kal_increment_is_variance (w₁ w₂ p : ℚ) :
    p * (1 - p) * (w₁ - w₂) ^ 2
      = p * (w₁ - meanFit w₁ w₂ p) ^ 2 + (1 - p) * (w₂ - meanFit w₁ w₂ p) ^ 2 := -- proof elided
```

3 The Lean formalization

Files and compile records. Everything is compiled with the pinned toolchain (Lean 4.34.1, Mathlib `d13f23b7`), through `scripts/certified_numerics/lean_pinned.py`, with `#print axioms` for every claimed declaration. Table 1 lists the files, their sha256 prefixes (full hashes in the ledger evidence), the exit code of the latest recompile and the number of `#print axioms` lines; each of the 11 compiles returned rc 0 with no errors (the `FixationDataCert.lean` row is the recompile after the doc-comment fix of this revision; the independent auditor recompiled the earlier text with the same axiom lines) (the producer transcripts of `Absorption` and `MoranModel` show deprecation warnings for `if_pos`/`if_neg`-style lemma names; the controls files show the expected `sorry` warning), and the axiom lines are identical to the producer transcripts. The claimed files contain no `sorry`, `admit`, `native_decide` or `axiom` token (whole-word grep over the claimed and check-only files; the planted controls are in separate files, and two doc-comment lines mention the words, `FixationDataCert.lean` line 15 and `MoranModel.lean` line 237).

file	role	sha256 (first 16)	rc	axiom lines
`Absorption.lean`	Absorption	`86fb5168a6b1efe0`	0	11
`WrightFisherFixation.lean`	WrightFisherFixation	`149a9328171a6347`	0	5
`MoranModel.lean`	MoranModel	`b0822dc0ee83505f`	0	16
`MoranFixation.lean`	MoranFixation	`c49806ac8b14295d`	0	8
`Price.lean`	Price	`1309c57a7818da57`	0	21
`FixationDataCert.lean`	check only	`87dea63cf4bd6492`	0	7
`PopGenProgImportAll.lean`	check only	`e26cdffa01ab893b`	0	6
`PriceImportAll.lean`	check only	`3e82333747ae4bdd`	0	4
`PopGenProgControls.lean`	planted, not imported	`685cdd545b9cff73`	0	2
`PriceControls.lean`	planted, not imported	`93bc16d08561f582`	0	2
`FixationDataCertControls.lean`	planted, not imported	`098c2ed751a65886`	0	2

Table 1: Lean sources (`formal_cert/Programmes/PopGen/`; module names are `Programmes.PopGen.`⟨role⟩). All compiles rc 0; the "axiom lines" are `#print axioms` outputs, each exactly `[propext, Classical.choice, Quot.sound]` for a claimed declaration; the two lines of each planted-controls file are the planted `sorry` and planted axiom, deliberately outside the allowlist.

Axiom footprint. Every claimed theorem prints `[propext, Classical.choice, Quot.sound]`. Example, verbatim from the `MoranFixation.lean` transcript:

```
'ProgPopGen.moran_fixation_limit' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgPopGen.moran_fixation_unique' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgPopGen.ctl_neg_wrong_limits' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Planted controls (caught). In a separate file that no claimed module imports, `ctl_sorry` is "proved" by `sorry` and `ctl_axiom` from a planted axiom; both claim the *wrong* limits $1/7$ and $1/3$ for the Moran chain at $N = 3$, $r = 2$, $i = 1$ (the kernel-checked limit is $4/7$). The file compiles with rc 0 and exit status is therefore not the gate; the axiom audit is [A, PG-A-0012]:

Listing 10: Planted controls (`PopGenProgControls.lean`, verbatim up to glyph rendering).

```
open ProgPopGen Filter Topology

theorem ctl_sorry :
    Tendsto (fun n => kPow (moranP 3 2) n 1 (Fin.last 3)) atTop (𝓝 (1 / 7)) := sorry

axiom ctl_cheat : False

theorem ctl_axiom :
    Tendsto (fun n => kPow (moranP 3 2) n 1 (Fin.last 3)) atTop (𝓝 (1 / 3)) := ctl_cheat.elim
```

```
'ProgPopGenControls.ctl_sorry' depends on axioms: [propext, sorryAx, Classical.choice, Quot.sound]
'ProgPopGenControls.ctl_axiom' depends on axioms: [propext, Classical.choice, ProgPopGenControls.ctl_cheat,
     Quot.sound]
```

Analogous planted pairs for the Price module and for the data-certificate module are in `PriceControls.lean` and `FixationDataCertControls.lean` (the independent auditor also planted a `sorry`, an axiom and a `sorry` hidden in a lemma of its own, all caught by `#print axioms`, and showed that changing the positive control $4/7$ to $1/3$ is rejected by the kernel) [A, PG-A-0012, PG-A-0014]; their transcripts show `sorryAx` and the planted `ctl_cheat` in the same way (Appendix transcripts `gen/axioms_controls_*.txt`).

Negative controls (kernel-checked, Tier A, PG-A-0008).

- `ctl_neg_identity_no_absorption`: the identity kernel on $\{0, 1, 2\}$ is stochastic and absorbing and $j/2$ is harmonic for it, yet $P^n(1, 2) = 0$ for every n, so it does not tend to $1/2$. The variance hypothesis of the main theorem cannot be dropped.
- `ctl_neg_flipped_value`, `ctl_neg_flipped_not_harmonic`: the exponent-flipped formula $(1 - r^i)/(1 - r^N)$ gives $1/7 \neq 4/7$ at $N = 3, r = 2, i = 1$ and violates the first-step equation.
- `ctl_neg_neutral_not_harmonic`: under selection, i/N violates the first-step equation.
- `ctl_neg_wrong_limits`: at chain level, $P^n(1, 3)$ for $N = 3$, $r = 2$ tends neither to $1/3$ nor to $1/7$ (it tends to $4/7$).
- Price: `ctl_neg_transmission_bias` ($\Delta\bar{w} = -1/2$ but $\mathrm{Var}/\bar{w} = 1/6$, while Price still holds, $-3/4 = -3/4$) and `ctl_neg_missing_wbar` ($\Delta\bar{w} = 1/6$ but $\mathrm{Var} = 1/4$) [A, PG-A-0010].

Listing 11: Negative controls (statements verbatim up to glyph rendering).

```
theorem ctl_neg_identity_no_absorption :
    InteriorHarmonic (idKernel 2) (fun j => ((j : ℕ) : ℝ) / 2) ∧
    (∀ n, kPow (idKernel 2) n 1 (Fin.last 2) = 0) ∧
    ¬ Tendsto (fun n => kPow (idKernel 2) n 1 (Fin.last 2)) atTop (𝓝 (1 / 2)) := -- proof elided

theorem ctl_neg_flipped_value : flipFix 3 2 1 = 1 / 7 ∧ flipFix 3 2 1 ≠ moranFix 3 2 1 := -- proof elided
```

```
theorem ctl_neg_flipped_not_harmonic :
    moranUp 3 2 1 * flipFix 3 2 2 + moranDown 3 2 1 * flipFix 3 2 0
      + (1 - moranUp 3 2 1 - moranDown 3 2 1) * flipFix 3 2 1 ≠ flipFix 3 2 1 := -- proof elided

theorem ctl_neg_neutral_not_harmonic :
    moranUp 3 2 1 * (2 / 3) + moranDown 3 2 1 * 0
      + (1 - moranUp 3 2 1 - moranDown 3 2 1) * (1 / 3) ≠ (1 / 3 : ℝ) := -- proof elided

theorem ctl_neg_wrong_limits :
    ¬ Tendsto (fun n => kPow (moranP 3 2) n 1 (Fin.last 3)) atTop (𝓝 (1 / 3)) ∧
    ¬ Tendsto (fun n => kPow (moranP 3 2) n 1 (Fin.last 3)) atTop (𝓝 (1 / 7)) := -- proof elided
```

Positive controls and non-vacuity (Tier A, PG-A-0009, PG-A-0010, PG-A-0011). Evaluated directly, not through the limit theorems: Wright–Fisher $N = 2$: $P^2(1, 2) = 3/8$ (not yet the limit $1/2$); Moran $N = 2$, $r = 2$: $p_1 = 1/3$, $q_1 = 1/6$ and $h(1) = 2/3$; $N = 3$, $r = 2$: $h = (0, 4/7, 6/7, 1)$ with the first-step equation holding; Price with three types: both sides equal $-13/36$. The hypotheses are satisfiable: `AbsorbingChain (moranP 3 2)` and `AbsorbingChain (wfP 3)` hold, and the earlier modules H11 and H12 import together with all P7 modules without name clashes.

4 Numerics and the rusty-SUNDIALS improvement (Tier X)

Method. The *uniformised* continuous-time chain has law $\pi(t) = \pi(0)\exp(t(P - I))$, the linear ODE $\dot{\pi} = \pi(P - I)$ with exact Jacobian $J_{jk} = P_{kj} - \delta_{jk}$. Since $\exp(t(P - I)) = \sum_n \mathrm{Poisson}(t; n)P^n$, the absorption mass $\pi_N(t)$ is a Poisson mixture of the n-step probabilities $P^n(i, N)$ and converges to the same limit [C, PG-C-0002]; this link is not formalised, so the numerics test the same closed forms but not the Lean matrix powers. The new crate `qf-popgen` of rusty-SUNDIALS (branch `physics/p7-fixation-2026-10-25`, commit `cda9a9057ec16820f2e16b560cc0e63cf7eb05b9`, whose parent `2b716c58fc7175798ad54dd9363938a3df37d9b0` is the commit the audit examined; base `7630f7e` = `origin/main`; not pushed) integrates this with CVODE (BDF, rtol 10^{-11}, atol 10^{-15}, analytic Jacobian), at output times $t = 1, 2, 4, \ldots$ until the interior mass is $\leq 10^{-9}$.

The improvement: a fail-closed certificate. `certify_absorption` returns a value only if *all* of the following hold, and otherwise an error (`InvalidKernel`, `Solver`, `NonFinite`, `MassDrift`, `NegativeMass`, `NotAbsorbed`, `ValueMismatch`), never a silent pass: the kernel is non-negative with unit row sums (10^{-12}) and absorbing ends; the state stays finite; $|\sum \pi - 1| \leq 10^{-9}$ at every output; *every component satisfies* $\pi_k(t) \geq -10^{-12}$ *at every output*; the interior mass has fallen below 10^{-9} by $t_{\max}$; and $|\pi_N - \mathrm{target}| \leq 10^{-8}$. The componentwise check is the improvement made in this revision: the audit observed that the total-mass and interior-mass tests are signed sums, so a state such as $(0.5, -0.25, 0.25, 0.5)$ (total mass 1, interior mass 0) would have passed them. It is now refused by `check_nonnegative` (unit test, Tier X); no reported run was affected, and the smallest component seen at any output of any accepted run is $-4.6e-18$. The binary `fixation_ode` exits with status 1 if any run is refused or any negative control is accepted. No change was made to LeanFlow in this lane, and no CVODE core code was altered; the contribution is a new workspace crate.

Runs. 78 Moran runs (the pairs (N, r): $(2, 2)$ and $(3, 2)$, which reproduce the Lean positive controls; $(10, 1.5)$, $(10, 0.8)$, $(10, 1)$, $(20, 1.2)$, $(30, 2)$) and 28 Wright–Fisher runs ($N = 10, 20$), each over every interior start $i = 1, \ldots, N - 1$: 106 accepted runs in all, targets S_i/S_N or i/N. Table 2 summarises them (the row `wf_neutral` pools $N = 10$ and $N = 20$); Figures 1–3 show the data.

run	N	r	starts	max |err|	max interior	max t_{stop}
`control_lean_N2_r2`	2	2	1	$6.6e-15$	$1.3e-14$	64
`control_lean_N3_r2`	3	2	2	$9.5e-14$	$1.2e-13$	128
`moran_N10_r1.5`	10	1.5	9	$1.0e-12$	$1.1e-12$	1024
`moran_N10_r0.8`	10	0.8	9	$4.1e-11$	$1.8e-10$	1024
`moran_N10_neutral`	10	1	9	$3.1e-10$	$6.3e-10$	2048
`moran_N20_r1.2`	20	1.2	19	$2.9e-12$	$3.3e-12$	4096
`moran_N30_r2`	30	2	29	$5.8e-13$	$5.8e-13$	2048
`wf_neutral`	10, 20	1	28	$5.3e-12$	$1.1e-11$	512

Table 2: CVODE absorption runs by case: the largest deviation from the closed form, the largest interior mass at the stopping time, and the longest stopping time. All are accepted by the certificate (tolerance 10^{-8} on the value, 10^{-9} on the interior mass).

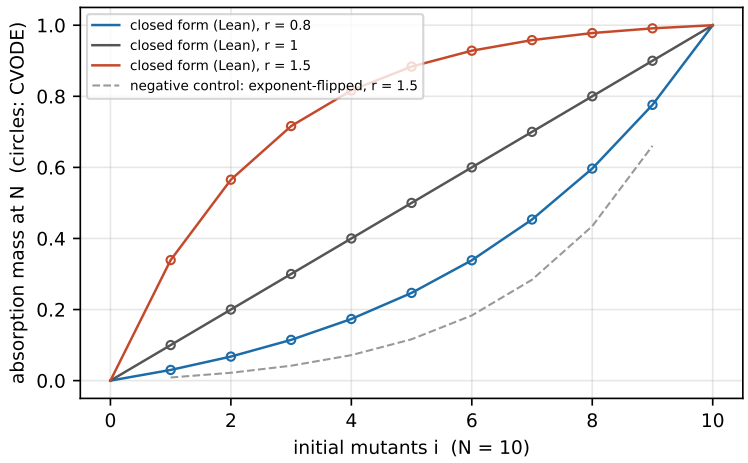


Figure 1: Moran, $N = 10$. Curves: S_i/S_N (the statement kernel-checked in Lean; evaluated here in floating point, Tier X). Circles: CVODE absorption mass. Dashed: the exponent-flipped negative control.

Controls. *Positive:* the Lean positive controls $(N, r, i) = (2, 2, 1) \mapsto 2/3$ and $(3, 2, \cdot) \mapsto (4/7, 6/7)$ are reproduced to $\leq 10^{-13}$; the neutral cases give i/N. *Negative* (all refused for the stated reason, none accepted; Table 3): the flipped exponent formula and the neutral i/N at $N = 10$, $r = 1.5$, $i = 1, 3, 7$ (`ValueMismatch`, the CVODE value being 0.339, 0.716, 0.958 against 0.0088, 0.042, 0.284 and 0.1, 0.3, 0.7); the identity kernel, a valid absorbing kernel with no absorption flow (`NotAbsorbed`; it mirrors the Lean negative control; it is not a Moran chain, so its fitness field in `results.jsonl` is `null`, where the first version of the file carried a meaningless $r = 1.5$); a horizon $t \leq 4$ (`NotAbsorbed`); a corrupted row sum (`InvalidKernel`); in addition a unit-level control for the new componentwise check (above). *Independent integrator:* the kernels rebuilt separately in `numpy` and $\exp(t_{\text{stop}}(P - I))$ from `scipy.linalg.expm` differ from the CVODE absorption mass by at most $1.0e-13$ over all runs [X, PG-X-0003]. The WORK1 cross-check with exact rationals and 60-digit matrix powers P^{4096} at $N = 10$ (errors $\leq 2 \times 10^{-61}$ for Wright–Fisher and $\leq 3.7 \times 10^{-50}$ for Moran $r = 3/2$; 6.6062×10^{-37} at most for $r = 1$ on the sampled starts) is recorded as [X, PG-X-0005].

Results. All 106 runs are accepted: largest deviation $3.1e-10$, largest interior mass $6.3e-10$, largest mass drift $9.8e-14$, smallest component $-4.6e-18$ [X, PG-X-0001]; all 9 negative controls are refused [X, PG-X-0002]. These numbers do not support any claim in this paper; they check that

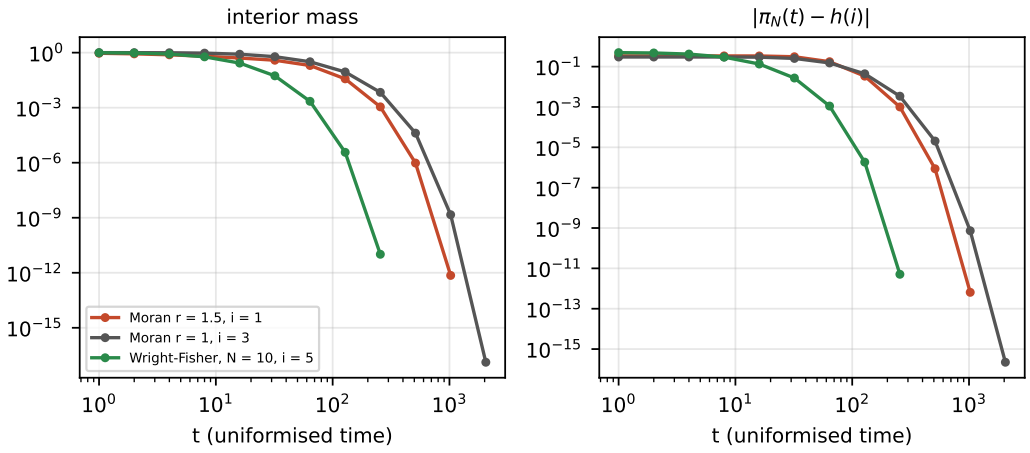


Figure 2: Interior mass and absorption error against time for three starts (CVODE, Tier X). The doubling output grid makes the geometric decay look like a log–log curve.

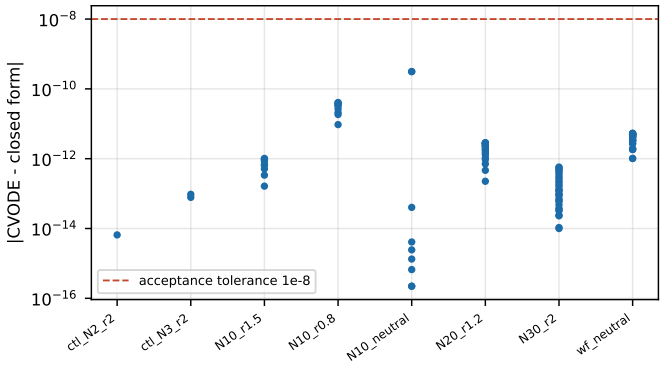


Figure 3: Deviation of the CVODE absorption mass from the closed form, all accepted runs, by case (Tier X).

an independent integrator and the Lean closed forms agree.

Data certificates: Lean checks the arithmetic of the decimal literals (Tier A + X). `FixationDataCert.lean` contains six theorems $|L - \texttt{moranFix}\ N\ r\ i| < 10^{-11}$ whose decimal literals L are the shortest round-trip decimal forms of the `p_fix` doubles written by `fixation_ode` (the JSONL file stores 18 significant digits, for example `6.666666666666660079e-1`, which round-trips to the literal `0.6666666666666601`), plus the negative control that the neutral literal 0.1 is *not* within 10^{-3} of `moranFix` 10 (3/2) 1 (Listing 12; [A, PG-A-0013]; provenance of the literals [X, PG-X-0004]). For the five Moran certificates the decimal literals differ from the exact value by 6.57×10^{-15}, 9.55×10^{-14}, 7.79×10^{-14}, 6.49×10^{-13} and 8.46×10^{-13} (exact rational arithmetic), and in the neutral case the literal `0.29999999999999977` differs from 3/10 by 2.30×10^{-16}; the float64 difference between `p_fix` and the double nearest 3/10, which is what the file's `abs_err` records, is 2.22×10^{-16}. All are far inside the proved bounds. The kernel certifies an inequality between a decimal literal and an exact rational, nothing more: it does not certify CVODE, and the literal is not a theorem about a Markov chain. A planted `sorry`/axiom pair for this module is caught [A, PG-A-0014].

control	start i	refused as
flipped_exponent_formula	1	ValueMismatch
neutral_i_over_N_under_selection	1	ValueMismatch
flipped_exponent_formula	3	ValueMismatch
neutral_i_over_N_under_selection	3	ValueMismatch
flipped_exponent_formula	7	ValueMismatch
neutral_i_over_N_under_selection	7	ValueMismatch
identity_kernel_no_absorption	1	NotAbsorbed
horizon_too_short	3	NotAbsorbed
corrupted_row_sum	3	InvalidKernel

Table 3: Negative controls of the certificate: 9 of 9 refused.

Listing 12: Data certificates (`FixationDataCert.lean`, three of the seven statements, verbatim up to glyph rendering; the two certificates and the negative control).

```
theorem cert_N3_r2_i1 : |(0.5714285714284759 : ℝ) - moranFix 3 2 1| < 1e-11 := -- proof elided

theorem cert_N10_r1p5_i1 : |(0.33921585523416337 : ℝ) - moranFix 10 (3 / 2) 1| < 1e-11 := -- proof elided

theorem ctl_neg_wrong_literal : ¬ (|(0.1 : ℝ) - moranFix 10 (3 / 2) 1| < 1e-3) := -- proof elided
```

Build hygiene. `cargo test -release -p qf-popgen - -test-threads=2` (commit `cda9a90`): 8 passed, 0 failed (positive: Moran $N = 3$, $r = 2$, neutral Wright–Fisher, a round-off-level state accepted by the componentwise check; negative: flipped and neutral targets, identity kernel, corrupted kernel, short horizon, cancelling-entries state refused by the componentwise check; plus a regression check that an accepted run records its minimum component, true by construction). The audit had run the earlier 6 tests at `2b716c5` (6 passed). `cargo clippy -workspace -features experimental --D warnings`: finished with no warnings; `cargo fmt -all -check`: rc 0 [X, PG-X-0007]. The re-run of `fixation_ode` at `cda9a90` gives 115 rows, 106 accepted, 9 refused, with every earlier field identical to the first version of the file; the only differences are the added `min_component` field and the `null` fitness of the identity-kernel control.

5 Limitations

- **No path measure.** The Tier A statements are limits of the matrix powers `kPow`. The identification with $\mathbb{P}_i(T_N \le n)$ is Tier C (PG-C-0001). The next step is to prove `kPow` $P\, n\, i\, j =$ (`Kernel.traj` $\kappa\ 0\ \delta_i$)$\{x \mid x_n = j\}$ with the Mathlib construction.
- **Moran variant.** The model is the one in which the individual that dies is drawn uniformly from all N; other variants (death-birth, graph structure) have different rates. The hypotheses are $0 < N$ and $0 < r$.
- **Not formalised:** times to absorption, the diffusion limit (Kimura), mutation, diploid selection, frequency-dependent fitness (here w is fixed within a generation), and Fisher's theorem in the partial-change reading (`fisher_general`, not achieved).
- **Uniqueness route.** For Moran, uniqueness is obtained from uniqueness of limits (Step 5), not from a maximum principle.
- **Independent AI recompile, but no human audit.** A separate AI auditor recompiled the modules from source and checked definition fidelity of `moranUp`/`moranDown`/`bdP` symbolically

against the Keller–Uğurlu rates (PG-C-0004) and the `Fin` indexing of `bdP`; this is an AI judgement from the producer's model family. No human audit exists: the ledger gate still flags every Tier A row for this reason, and the `audit` fields of the ledger stay null (Section 7). The auditor did not see the post-audit changes (the doc comment and the componentwise check).

- **Sources.** Moran (1958), Price (1970) and Price (1972) were *not* read; they appear only as listed in [1] (p. 18) and [3] (p. 69). The novelty search was a few `alphaXiv` queries; absence by search is not proof, and no priority claim is made.
- **Genetic crosswalk** is textual (PG-C-0007) and the Price module has not been built inside that repository.
- **Numerics** are Tier X, test the uniformised ODE rather than the Lean matrix powers, and used $N \leq 30$.

6 Reproducibility

- **Lean.** Files under `formal_cert/Programmes/PopGen/` of the AutoevolveAI worktree `openai-math-discovery` (uncommitted at the time of writing; base commit `9d24574`). Full sha256 values are in `ledger/evidence/`. The independent audit is `AUDIT.md` and the list of changes made in answer to it is `REVISION.md`, both in the paper directory. Compile: from the repository root, `taskset -c 4-7 python3 scripts/certified_numerics/lean_pinned.py compile FILE -json-out OUT` (pinned Lean 4.34.1 and Mathlib; order: Absorption, WrightFisherFixation, MoranModel, MoranFixation, Price, then the import-all and controls files; Absorption, MoranModel, MoranFixation and WrightFisherFixation were also built as importable `.olean` modules).
- **Rust.** `CARGO_TARGET_DIR=... cargo build -release -j 2 -offline -p qf-popgen`, then `fixation_ode OUT_DIR` writes `results.jsonl` and `summary.json` (copied to `data/`). Branch `physics/p7-fixation-2026-10-25`, commit `cda9a9057ec16820f2e16b560cc0e63cf7eb05b9` (parent `2b716c58...`, audited); not pushed.
- **Figures and tables.** `gen/gen_all.py` (figures, run table, independent `expm` check), `gen/gen_listings.py` (verbatim Lean excerpts and axiom transcripts from the hashed sources). Run with the repository `.venv`.
- **Earlier work.** Same tier discipline and ledger gate as the companion preprints (DOIs 10.5281/zenodo.23646103, 23646105, 23646107, 23646111, 23646117, which were only reserved when this version was written and resolve once those deposits are published) and the universal-optimality textbook (doi:10.5281/zenodo.23295142); the P7 results do not use their theorems.

7 Ledger and gate

The ledger has 36 rows: 14 Tier A, 5 Tier L, 8 Tier C, 9 Tier X; every row has a content-addressed evidence blob. Tier A rows depend only on Tier A rows; Tier C and X rows are never cited as Tier A support. The Elenchus gate (`tools/ledger.py ledger.json -evidence-dir evidence`) reports **0 blocking findings and 14 flags, exit code 1**: each flag is `LEDGER_UNAUDITED_TIER_A`, one per Tier A row whose statement adequacy no person has audited (verbatim output in `ledger/gate_output.txt`). Exit code 1 is the expected outcome while the audit field is null; it is not hidden. The independent AI recompile is recorded as the Tier X row PG-X-0009 and does not fill any `audit` field, since it is not a human statement audit.

rows	content
PG-A-0001..0004	absorption limit and uniqueness (general); Wright–Fisher; Moran lemmas; Moran chain theorems
PG-A-0005..0007	Price equation; haploid increment; Genetic-repo crosswalk (textual)
PG-A-0008..0011	negative controls; positive controls; Price controls; non-vacuity and import-together
PG-A-0012..0014	planted controls caught (fixation, Price); data-certificate arithmetic; its planted controls
PG-L-0001..0005	Keller–Uğurlu rates and fixation formulas; Marion (`Kernel.traj`); LaGatta Price/Fisher; Price (1972) via LaGatta and Baez
PG-C-0001..0008	bridge to probability; Poisson link; human proofs; Moran definition fidelity; Price scope (`fisher_general` not achieved); no novelty; Genetic note; no openai/math dependence
PG-X-0001..0009	CVODE runs; refused controls; `expm` check; literal provenance; exact/mpmath; Price numerics; build hygiene; paper-stage recompile (same pipeline); independent AI recompile and audit

Table 4: Ledger rows.

References

[1] P. Keller and M. Uğurlu, *Fixation probability in Moran-like Processes on graphs*, arXiv:2403.12598v2 [math.PR], 11 Sep. 2024 (eqs. (3), (4), pp. 2–4; source typos on pp. 2, 3 and 7 not copied; [Mor58] on p. 18).

[2] E. Marion, *A Formalization of the Ionescu-Tulcea Theorem in Mathlib*, arXiv:2506.18616v5 [math.PR], 17 Mar. 2026 (pp. 5, 9, 12).

[3] T. LaGatta, *The Mathematics of Evolution: The Price Equation, Natural Selection, and Environmental Change*, arXiv:2202.10289v4 [math.PR], 4 Dec. 2022 (eqs. (1.6), (1.8), (2.24), (2.27), pp. 9, 18, 19; reference list p. 69).

[4] J. C. Baez, *The Fundamental Theorem of Natural Selection*, arXiv:2107.05610v3 [q-bio.PE], 6 Oct. 2021 (pp. 1, 2, 4).

[5] P. A. P. Moran, Random processes in genetics, *Math. Proc. Cambridge Philos. Soc.* **54** (1958) 60–71. *As cited in [1]; not read.*

[6] G. R. Price, Selection and covariance, *Nature* **227** (1970) 520–521. *As listed in [3], p. 69; not read.*

[7] G. R. Price, Fisher's "fundamental theorem" made clear, *Ann. Hum. Genet.* **36**(2) (1972) 129–140. *As listed in [3], p. 69; not read.*

[8] Earlier modules `PopGenWrightFisher.lean` (H11) and `PopGenFisher.lean` (H12), AutoevolveAI `formal_cert/Autoresearch/`, results of 2026-10-25; independently recompiled by a separate AI verifier (ACCEPT_WITH_CORRECTIONS), no human audit.